

Decision-Making Needs Expression

AI-Based Intelligent System for Network Security Monitoring & Automated Response
Integrating Port Scanning Detection, Moving Target Defense (MTD) & Honeypot Mechanisms

Module : Artificial Intelligence | Status : Academic Project

Module	Artificial Intelligence	Attack Focus	Port Scanning & Network Reconnaissance
Status	Academic Project	Severity Level	MEDIUM-HIGH

Project Team				
El Kartouti Anas	Lekhbioui Adil	Moulay Anas	Hammoubel El Yazid	Nafia Khadija
Member	Member	Member	Member	Member

1. Executive Summary

This document presents the formal decision-making needs expression for the design, implementation, testing, and validation of an Artificial Intelligence model dedicated to the real-time detection and automated mitigation of **Port Scanning and Network Reconnaissance** activity within networked environments. Reconnaissance constitutes the first phase of virtually every cyberattack kill chain: an adversary probes the network to map active hosts, enumerate open ports, and fingerprint running services before launching a targeted intrusion. Early detection and containment at this phase can prevent the full materialization of a subsequent, more destructive attack.

Beyond detection, this project integrates two advanced and complementary proactive defense mechanisms: **Moving Target Defense (MTD)**, which continuously mutates the network's observable attack surface to disrupt and mislead scanning activity, and **Honeypots**, which deploy decoy assets to attract, trap, and profile adversarial reconnaissance behavior. Together, these three pillars — detection, deception, and surface mutation — form a layered, intelligent security posture capable of addressing reconnaissance threats with significantly greater depth than any single-layer approach.

2. Problem Statement

Port scanning, in its various forms — SYN scan, ACK scan, UDP scan, XMAS scan, FIN scan — generates traffic patterns that differ fundamentally from legitimate host-to-host communication. An adversary conducting a scan sends probes to hundreds or thousands of ports in succession, producing an anomalous distribution of half-open connections, RST responses, and unanswered packets.

Despite being a well-understood technique, **detecting stealthy or slow-rate scanning** — where the adversary deliberately spaces probes to evade threshold-based detection — remains a significant challenge. Such low-and-slow patterns require behavioral analysis over extended time windows rather

than instantaneous packet-level inspection.

A further and critical dimension of the problem is the **static, predictable nature of conventional network configurations**. Fixed IP addresses, stable port assignments, and consistent service layouts give adversaries a reliable map to exploit. Once a scan is complete, the attacker possesses a stable, actionable picture of the infrastructure — a problem that no detection system alone can fully address. Additionally, once scanning activity is detected, organizations currently lack systematic means to **observe and profile** the adversary's intent and methodology in a controlled, risk-free manner.

■ 3. Project Objectives

Primary Objective	Train and deploy an AI model capable of detecting port scanning and network reconnaissance activity in real time across networked environments.
Scan Type Coverage	Detect and classify SYN Scan (stealth), TCP Connect Scan, UDP Scan, ACK Scan, XMAS Scan, and slow/distributed scanning variants.
Moving Target Defense	Integrate an MTD mechanism that continuously rotates network configuration parameters — including port assignments and IP addresses — to invalidate adversarial reconnaissance maps and increase the cost of scanning.
Honeypot Deception	Deploy honeypot nodes that mimic legitimate services to attract, trap, and observe adversarial scanning activity, enriching threat intelligence and enabling deep attacker behavioral profiling.
Attacker Profiling	Automatically build a structured profile for each detected threat actor: network identity, scan methodology, targeted scope, behavioral timeline, and interaction history with decoy assets.
Alerting	Trigger real-time, graduated alerts upon detection of scanning activity, classified by scan type, confidence score, and severity level.
Automated Response	Enforce automated protective measures — including network-level blocking — upon confirmed threat detection, without requiring human intervention.

■ 4. Functional Requirements

4.1 System Inputs

- Raw TCP/UDP packets captured on the network interface, filtered for partial or incomplete handshake patterns indicative of probing behavior
- Labeled training dataset containing port scanning and benign traffic samples (CICIDS2017 PortScan subset, UNSW-NB15 Reconnaissance category)
- Sliding time-window aggregated flow features computed per source IP over configurable observation windows
- Network metadata: source/destination IP, MAC addresses, ports, protocol, TCP flag combinations, TTL values, and timestamps
- Honeypot interaction logs: connection attempts, payloads, source identifiers, and session metadata from decoy nodes
- MTD state feed: current port assignment table and IP rotation schedule, enabling the AI engine to correctly interpret traffic against the dynamic topology

4.2 System Outputs

- Real-time dashboard: per-source port probe counts, scan heat map, active threat list, honeypot interaction events, MTD rotation log
- Per-time-window classification: BENIGN or PORT_SCAN, with inferred scan type and associated confidence score
- Attacker profile card: source IP and MAC, scan type, ports and hosts targeted, first/last seen timestamps, honeypot interaction flag, threat level
- Graduated alert notification with scan type label, severity score, and contextual metadata
- Automated network-level block rule applied to the confirmed attacker's source address, with the event logged to the incident database
- MTD rotation trigger: upon detection, the system initiates a port/IP rotation cycle to further invalidate any reconnaissance data collected by the attacker

■ 5. Dataset Specification

Dataset	Relevant Subset	Samples (approx.)	Source
CICIDS2017	PortScan category	~158,000 flows	UNB / CIC
UNSW-NB15	Reconnaissance category	~13,000 records	UNSW Australia
CAIDA Telescope	Backscatter / SYN flood probes	Variable captures	CAIDA.org

■ 6. Key Network Features for the ML Model

Feature Name	Description	Relevance to Port Scanning
Distinct Dst Ports/IP	# unique destination ports contacted by source	Critical — core scanning signature
SYN Flag Count	SYN packets sent without completing handshake	Critical — stealth SYN scan
RST Flag Count	RST responses from closed ports	Very High — port probe rejection
Flow Duration	Duration of each individual probe connection	High — very short in fast scans
Total Fwd Packets	Packets per flow from scanner	High — usually 1 per probe
ACK Flag Count	ACK packets without prior SYN	Medium — ACK scan detection
IAT Mean	Mean inter-arrival time between probes	High — slow scans have large IAT
Unique Dst IPs/Src	# distinct destination hosts contacted	Medium — host discovery phase
TTL Value	Time-to-live field in IP header	Low — OS fingerprinting attempts
Bwd Packet Length	Response packet size	Medium — RST vs SYN-ACK ratio

Feature Name	Description	Relevance to Port Scanning
Honeypot Interaction Flag	Binary: did source IP contact a decoy node	Critical — high-confidence threat signal
MTD Port Delta	Difference between probed port and current assigned port	High — reveals stale reconnaissance data

■ 7. Machine Learning Model Strategy

Primary Model	Random Forest Classifier — handles high-dimensional scan features with natural feature importance ranking and robustness to class imbalance
Secondary Model	XGBoost — compared for accuracy on time-windowed aggregated features and as a high-performance ensemble benchmark
Slow Scan Detection	Isolation Forest (unsupervised) applied on sliding 60-second window aggregates per source IP to capture low-and-slow probing patterns
Feature Engineering	Time-window aggregation: distinct ports/IP/10s, SYN-ratio, IAT statistics, honeypot interaction count, and MTD port delta over rolling windows
Imbalance Handling	SMOTE oversampling on the PORT_SCAN minority class in the training set
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, AUC-ROC, Detection Rate per scan type, Honeypot True Positive Rate
Validation Strategy	Stratified K-Fold (k=10) + temporal holdout (last 20% by time order)

■ 8. MTD & Honeypot Component Specifications

8.1 Moving Target Defense (MTD)

The MTD component continuously mutates key network configuration parameters to invalidate any reconnaissance data an adversary may have collected. The core principle is to impose a high and continuous cost on the attacker: a network map obtained through scanning becomes stale within minutes, forcing repeated re-scanning that is itself detectable.

Mutation Scope	Dynamic rotation of service port assignments and, optionally, virtual IP addresses of protected hosts within the network
Rotation Trigger	Time-based (scheduled rotation interval) and event-based (triggered automatically upon confirmed scanning detection by the AI engine)
AI Integration	The MTD state table is fed as a live input to the AI model, ensuring that traffic directed at recently vacated ports is correctly identified as post-scan probing rather than legitimate communication
Coordination	All legitimate nodes on the network are synchronized with the current MTD state via a secure distribution mechanism, ensuring zero disruption to authorized traffic during rotation cycles

8.2 Honeypot Integration

Honeypot nodes are strategically deployed within the network topology as high-fidelity decoy systems. They present a realistic appearance to any scanning adversary while containing no operational data or functionality. All traffic directed to a honeypot is inherently unauthorized and constitutes an unambiguous

threat signal.

Honeypot Type	Low-to-medium interaction honeypots emulating common services (SSH, HTTP, FTP) to attract and engage scanning and post-scan probing activity
Placement Strategy	Distributed across the network topology to maximize the probability of contact during any broad or targeted scanning campaign
AI Contribution	Honeypot interaction events — connection attempts, payloads, source IPs — are ingested as real-time features by the AI model, significantly boosting detection confidence for confirmed threat actors
Profiling Output	Each honeypot interaction is logged and associated with the corresponding attacker profile, enriching the threat intelligence record with behavioral indicators, service targeting patterns, and session-level metadata

■ 9. Non-Functional Requirements

Requirement	Criterion	Target Value
Fast Scan Detection	Detection time for aggressive Nmap scan	< 8 seconds
Slow Scan Detection	Detection time for stealthy / slow scan	< 90 seconds (sliding window)
Model F1-Score	On CICIDS2017 PortScan test subset	≥ 88%
Model Accuracy	Overall classification on test set	≥ 90%
False Positive Rate	Normal connections flagged as scans	< 6%
Dashboard Refresh	Scanner activity & honeypot heat map update frequency	≤ 5 seconds
Blocking Response	Time from detection to automated block rule enforcement	< 15 seconds
MTD Rotation Latency	Time from detection trigger to completed port rotation	< 30 seconds
Honeypot Detection Rate	Proportion of scanning IPs that interact with at least one honeypot	≥ 60%

■ 10. Project Phases & Deliverables

Phase	Activities	Key Deliverables
1 — Design	Network topology design (multi-host with honeypot nodes), MTD architecture, feature set definition, scan type taxonomy	Network diagram, MTD design spec, feature dictionary
2 — Implementation	Dataset preprocessing, RF + XGBoost + Isolation Forest training, real-time capture pipeline, MTD rotation engine, honeypot deployment, dashboard	Trained models (.pkl), IDS pipeline, MTD engine, honeypot nodes, dashboard UI

Phase	Activities	Key Deliverables
3 — Testing	Simulate Nmap SYN, Connect, UDP, XMAS, slow scans; validate MTD disruption effectiveness; test honeypot engagement	Per-scan detection table, ROC curves, MTD effectiveness report, honeypot engagement log
4 — Validation	K-Fold cross-validation, temporal holdout, end-to-end system validation including MTD and honeypot integration	Final report, demo recording, multi-model comparison table

■ 11. Technical Stack Summary

OS / Virtualization	Ubuntu Server (IDS + victim nodes), Kali Linux (attacker), GNS3 / VirtualBox
Programming Language	Python 3.10+
Traffic Capture	Scapy with BPF filters (SYN-only), PyShark, sliding window aggregation
ML Libraries	Scikit-learn, XGBoost, imbalanced-learn, Joblib, Pandas, NumPy
Web Dashboard	Flask + Socket.IO + D3.js (port heat map, honeypot event feed) / Grafana + InfluxDB
Blocking Engine	Linux iptables DROP rules via Python subprocess, with TTL-based auto-unblock scheduler
MTD Engine	Custom Python service managing dynamic port/IP rotation table, state distribution to legitimate nodes, and AI engine synchronization
Honeypot Framework	Cowrie (SSH/Telnet), Dionaea (multi-service), or custom lightweight Python socket listeners — integrated with the AI ingestion pipeline
Attack Simulation	Nmap (SYN, ACK, UDP, XMAS, slow scans), Masscan, custom Scapy scripts
Version Control	Git + GitHub — feature branches per module component

■ 12. Expected Outcomes & Success Criteria

- ✓ The trained model achieves an F1-Score $\geq 88\%$ on the CICIDS2017 PortScan test subset.
- ✓ An aggressive Nmap SYN stealth scan is detected and an alert is triggered within 8 seconds.
- ✓ A slow-rate scan (1 probe/second) is correctly identified within the 90-second analysis window.
- ✓ The MTD engine completes a port rotation cycle within 30 seconds of a confirmed detection trigger.
- ✓ Following MTD rotation, continued probing of vacated ports by the same source IP is correctly classified as post-scan activity with no false negative.
- ✓ At least 60% of scanning source IPs interact with at least one honeypot node during a full scan campaign, generating enriched attacker profile data.
- ✓ The dashboard reflects scanning events, honeypot interactions, and MTD rotation status with a maximum update delay of 5 seconds.
- ✓ Automated blocking is enforced within 15 seconds of a confirmed detection.
- ✓ The attacker profile correctly identifies scan type (SYN / UDP / ACK / XMAS) and includes honeypot contact history where applicable.

✓ Legitimate traffic from authorized nodes generates a false positive rate below 6% across all test scenarios, including during active MTD rotation cycles.
